

Engineered *Vibrio natriegens* lysate can replace multiple components of cell culture media

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Abstract

Reducing the cost and environmental impact of cell culture media is an important goal for cultivated meat, the process of generating meat *in vitro* using proliferating animal cells. While prior approaches have demonstrated the use of microbial lysates to replace expensive animal-based fetal bovine serum (FBS) in media, these formulations still rely on large quantities of growth factors such as fibroblast-like growth factor 2 (FGF2). Here, we demonstrate the use of FGF2-expressing *Vibrio natriegens* to create whole-cell lysates that replace both FBS and FGF2 in cell culture media for cultivated meat applications. This medium, named “VN40^{FGF}”, supports rapid proliferation of immortalized bovine muscle satellite cells (iBSCs) in the absence of supplemented FGF2. Cells grown in VN40^{FGF} maintain phenotype and differentiation capacity. We also demonstrate that engineered *V. natriegens* can grow in spent cell culture media, further improving sustainability and economics, and reducing potential eutrophication concerns associated with waste disposal. Our approach combines multiple strategies for reducing the total number of media inputs, demonstrating opportunities for more economical and sustainable cell culture, especially for cultivated meats.

Keywords: bovine cells; serum; lysate; hydrolysate; recombinant proteins; cultivated meat; cellular agriculture

Introduction

Cultivated meat, the process of generating meat using *in vitro* culture of animal cells, is a promising alternative to conventional livestock agriculture. Benefits of cultivated meat include streamlining food production, increasing protein availability, improving food safety, and reducing the negative externalities associated with livestock.^{1,2} However, cultivated meat has not reached significant market penetration due to its high production costs. Raw ingredient costs remain a significant roadblock to making cultivated meat reasonably accessible and affordable.

Cell culture media represents the most expensive input ingredient in cultivated meat production, comprising 98 % of raw material costs at scale.³ Recombinant growth factors are the primary driver of media costs, with some estimates attributing 95 % of serum-free media cost to growth factors alone.⁴ Eliminating the costly animal-derived fetal bovine serum (FBS) ingredient from cell culture media, while an important goal for cultivated meat, leads to manyfold increases in growth factor requirements.⁵ For example, immortalized bovine satellite muscle cells (iBSCs) require supplementation with numerous growth factors, including neuregulin, transforming growth factor (TGF), and fibroblast growth factor 2 (FGF2) in order to grow in the absence of FBS.⁶ These growth factors cost thousands of USD per milligram and thus impart significant costs to serum-free media (SFM). FGF2 alone, which is required at particularly high concentrations in SFM, is responsible for more than 50 % of BSC media costs.⁷ Growth factor production also carries significant negative environmental impacts, such as increases to global warming potential, ecotoxicity, and eutrophication risks.⁸ The costs of growth factors, both economic and environmental, currently serve as a significant obstacle to the low-cost sustainable production of cultivated meat.

Growth factor costs are particularly high due to their labor intensive and low-yield production process, in which they are expressed as recombinant products, typically in *E. coli* or mammalian hosts, and purified by affinity chromatography.^{9,7} Efforts to reduce growth factor costs have primarily focused on increasing yields through methods such as enhanced expression (via fusion tags, strain engineering, and metabolic optimization),¹⁰ bioprocess improvement,¹¹ and optimization of purification.¹² However, these innovations only marginally reduce growth factor costs. The consumables and labor associated with growth factor purification account for around 94 % of final costs.⁹ Growth factors have also been produced in spent mammalian cell culture media, but only used from these processes in their purified form.^{13,14} To avoid using purified growth factors in SFM, whole-cell lysates from growth factor-expressing microbes can be used instead. While this has been attempted previously using *Saccharomyces cerevisiae* yeast expressing a combination of four cytokines,¹⁵ it was not used with SFM. Our group's previous work successfully formulated

SFM – substituting recombinant albumin with lysates from the rapidly-growing bacterium, *Vibrio natriegens*.¹⁶ In the work, we demonstrated that *V. natriegens* lysates stimulated the growth of iBSCs at rates superior to that of a more expensive previously established Beefy-9 SFM.⁶ This established a low-cost easy to produce medium called “VN40” for its use of *V. natriegens* lysate at 40 µg protein·mL⁻¹. The VN40 medium was advantageous in that it did not require large amounts of expensive purified recombinant protein, as in Beefy-9 and other studies.^{6,9,17-18} The lysate required only steps of cell growth, lysis, and filtration, making it easier to prepare than other SFM which require laborious protein extraction from plant matter or relatively slow-growing algae.¹⁹⁻²¹ VN is the fastest growing organism known, with a doubling time of about 10 min,²² and stimulates BSC growth at a low concentration of 40 µg·mL⁻¹, making overall production inexpensive and rapid. However, like other cultivated meat SFM, the VN40 media still requires a relatively high concentration of supplemental commercial FGF2 of human origin. At a concentration of 40 ng·mL⁻¹, FGF2 adds \$ 36 L⁻¹ to VN40 media costs (laboratory-scale). Fortunately, VN is a genetically tractable organism with established use in recombinant protein production.²³ Recently, *V. natriegens* NC7 strain was developed for low resource-intensity recombinant expression, demonstrating a novel economical and highly efficient transformation method.²⁴ Thus, we endeavored to engineer the *V. natriegens* NC7 strain to generate a lysate-based SFM that could be used to replace growth factors for cultivated meat production.

In this work, we demonstrate the production of a *V. natriegens* lysate containing expressed bovine FGF2 and successfully replaced both serum and FGF2 in BSC medium. This led to the development of VN40^{FGF}, a serum-free medium, containing no additional FGF2. VN40^{FGF} showed comparable cell growth and phenotypic outcomes to previously developed VN40 medium, which was supplemented with human FGF2 at 40 ng·mL⁻¹. We further show that the lysate can be prepared from spent mammalian cell culture media, demonstrating a closed-loop production process to further reduce production costs. In all, this study demonstrates a significant cost reduction for the process of generating cultivated meat, unlocking potential further investigations into using whole-cell lysates engineered to replace other expensive growth factors and cytokines. Overall, these findings prompt advances in cultivated meat towards an affordable reality for the general population.

Materials & Methods

V. natriegens transformation

A single colony of *Vibrio natriegens* NC7 (Addgene bacterial strain #215356) was used to inoculate 20 mL Minimal Competence Medium (MCM, 9 mM HEPES pH 7.4, 2 mM sodium acetate, salts). The culture was incubated at 30 °C, static, for 20 h. 25 ng of plasmid DNA, either empty vector or containing the bovine FGF2 (bFGF2) insert, was added to 350 µL culture, then incubated in static culture for 30 min (**Figure S1**). The transformation mixture was then plated directly on Lysogeny Broth (LB) + 2 % NaCl + 1.5 % agar + 2 µg·mL⁻¹ chloramphenicol, then incubated at 37 °C for 1–2 days until colonies appeared.

Extraction of microbial lysate

A single colony of *Vibrio natriegens* NC7 transformed with the bFGF2 plasmid was used to inoculate 5 mL LB + 2% NaCl + 2 µg·mL⁻¹ chloramphenicol. The culture was incubated overnight at 30 °C, shaking at 200 rpm. The overnight culture was diluted 1:100 into 50 mL LB + 2 % NaCl + 2 µg·mL⁻¹ chloramphenicol, then grown at 30 °C until the OD₆₀₀ reached 0.6–0.9. At this point, 50 µL IPTG (isopropyl β-D-1-thiogalactopyranoside) was added to the culture, and incubation continued for 4–6 h. The cell pellet was harvested by centrifuging the culture at 5,000 × g for 30 min, washing and resuspending in 15 mL of sterile phosphate buffer saline (PBS, 10 mM phosphate, 140 mM NaCl, 3 mM KCl, pH 7.4). To generate lysate, the cell suspension was sonicated (Branson 150 sonicator and 10 s Pulse, 30 s gap between each pulse and a total 7 min at 45 % amplitude) on ice. After sonication, the soluble lysate was obtained by centrifuging the sonicated solution at 21,000 × g for 20 min. The supernatant was then filtered through a 0.2 µm syringe filter (Corning #431219). The total protein content of the lysate was determined using Bradford Assay (Coomassie Plus™ Protein Assay Reagent, ThermoFisher #23238), using bovine serum albumin (BSA) to generate a standard curve. The concentration of FGF2 in the lysate was determined using the Invitrogen Bovine/Human bFGF ELISA Kit (ThermoFisher #EB2RB).

iBSC culture and growth studies

iBSCs were previously isolated from the semitendinosus of a 5-week-old Simmental calf at the Tufts Cummings School of Veterinary Medicine.⁶ These cells were engineered to constitutively express bovine telomerase reverse transcriptase (TERT) and cyclin-dependent kinase 4 (CDK4) in prior studies.²⁵ After adaptation to VN40 media as described previously,¹⁶ iBSCs at passage 70 were seeded in triplicate at 2,000 cells·cm⁻² in serum-free, FGF2-free B8 medium with 1.5 µg·cm⁻² vitronectin (rhVTN-N; ThermoFisher #A31804) in tissue culture polystyrene 6-well plates. B8 medium was formulated as previously described,⁶ omitting FGF2. After allowing iBSC adherence for at least 3 h, cells were washed with DPBS (Dulbecco's phosphate buffered saline; ThermoFisher #14040117) and fed B8

supplemented either human FGF2 (hFGF2) at 40 ng·mL⁻¹ (PeproTech #100-18B, Rocky Hill, NJ, USA) or 40 µg·mL⁻¹ lysate from bFGF2-expressing *V. natriegens*. Negative controls were fed B8 with no added FGF2. Cells were fed every 2–3 days until they reached 80–90 % confluence. At this point, cells were passaged using TrypLE Express enzyme (ThermoFisher #12604013) and counted using a NucleoCounter NC-200 (Chemometec, Allerød, Denmark). Doubling time (*DT*) was calculated using **Equation 1**

$$DT = \frac{\Delta t}{\log_2\left(\frac{C_f}{C_i}\right)} \quad \text{(Equation 1)}$$

with Δt being time between passages, C_f being cell count at passage, and C_i being total cells seeded. At passage, cells were centrifuged at 300 × *g* and seeded in a new well. Cells were cultured at 37 °C and 5 % CO₂. During feeding of iBSCs with VN40^{FGF}, spent media was collected for growth of *V. natriegens* for recycled media studies. For these studies, cells were similarly fed VN40^{FGF} produced from spent media and grown and counted as described.

Characterization and differentiation of iBSCs

After the fifth passage in VN40/VN40^{FGF} media, iBSCs were allowed to adhere in triplicate in 48-well plates at 6000 cells·cm⁻² and their satellite cell identity was characterized via staining for Paired-box 7 (Pax7). Cells were fixed with 4 % paraformaldehyde in PBS (ThermoFisher #AAJ61899AK) for 20 min, washed with DPBS containing 0.1 % Tween-20 (ThermoFisher, BP337), permeabilized for 10 min with DPBS containing 0.1 % Triton X-100 (ThermoFisher, BP151), washed again with DPBS + Tween-20, and blocked for 30 min using a blocking solution containing DPBS with 5 % goat serum (ThermoFisher #16210064) and 0.05 % sodium azide (Sigma #S2002). A primary antibody solution containing 1:500 Pax7 antibody (ThermoFisher #PA5-68506) and 1:1000 actin stain Phalloidin 488 (ThermoFisher A12379) in blocking solution was added to cells and incubated overnight at 4 °C. Cells were then washed with DPBS + Tween-20 and blocked for 30 min once again. A solution containing 1:500 Pax7 secondary antibody (ThermoFisher #A11072) and 1:1000 nuclear stain DAPI (ThermoFisher #D1306) in blocking solution was added to cells and incubated for 1 h at room temperature. Cells were then washed with DPBS + Tween-20 and kept in DPBS for imaging. Imaging was carried out using fluorescence microscopy (KEYENCE, BZ-X700, Osaka, Japan). Stemness was quantified using a Celigo automated image cytometer (Revvity, Waltham, MA) to determine cells positive for Pax7. Cells were differentiated by seeding at 5,000 cells·cm⁻² in 48-well plates and initiating differentiation when cells appeared confluent 3 days after initial seeding. Differentiation media contained DMEM + Glutamax supplemented with 2 % horse serum (ThermoFisher #16050130), 0.5 mg·mL⁻¹ recombinant human albumin, 1 × ITS-X (ThermoFisher #51400056), 0.5 mM LDN193189 (Sigma #SML0559), 1 % antibiotic/antimycotic, and 2.5 µg·mL⁻¹ puromycin. Differentiation continued for 9 days with

cells fed fresh differentiation media every 2–3 days. After mature myotubes were observed, cells were fixed, permeabilized, and blocked as described. Cells were then stained using myosin heavy chain (MHC; Developmental studies hybridoma bank #MF-20, Iowa City, IA, USA) and 1:200 desmin (Abcam #ab15200, Cambridge, UK) primary antibodies diluted in blocking solution. Secondary antibodies for MHC (ThermoFisher #A32723; 1:200) and desmin (ThermoFisher #A11072; 1:500) along with 1:1,000 DAPI, diluted in blocking buffer, were then applied as described. Imaging was carried out using fluorescence microscopy (KEYENCE, BZ-X700, Osaka, Japan). Differentiation was quantified using Celigo automated image cytometer to determine percentage of cells contained in desmin-positive myotubes (fusion index).

Spent medium recycling

To generate VN40^{FGF} from spent cell culture media, we followed the scheme in **Figure 1**. The spent medium from iBSC cell culture was collected, supplemented with 3 % NaCl (w/v), and autoclaved with a 40 min liquid cycle. *V. natriegens* NC7 was cultured overnight in LB + 2 % NaCl, then subcultured into the sterilized spent media at a starting OD₆₀₀ of 0.05. Growth curves were generated using the BioTek Epoch2 Microplate Spectrophotometer (Agilent Technologies, Santa Clara, CA, USA), taking new OD₆₀₀ measurements every 10 min for 24 h. Spent media collected from iBSC and *V. natriegens* growth was analyzed for ammonium, glucose, glutamate, glutamine, lactate, and potassium using YSI 2950 Biochemistry Analyzer (Xylem Corp., Washington, DC), using buffer standards and sensors provided by the manufacturer.

Statistics

Statistical analysis was carried out using GraphPad Prism v10.6. Significant measurements were carried out using Student's t-test, or one-way ANOVA with Dunnett's correction for multiple comparisons to detect the presence of significantly different groups.

Results

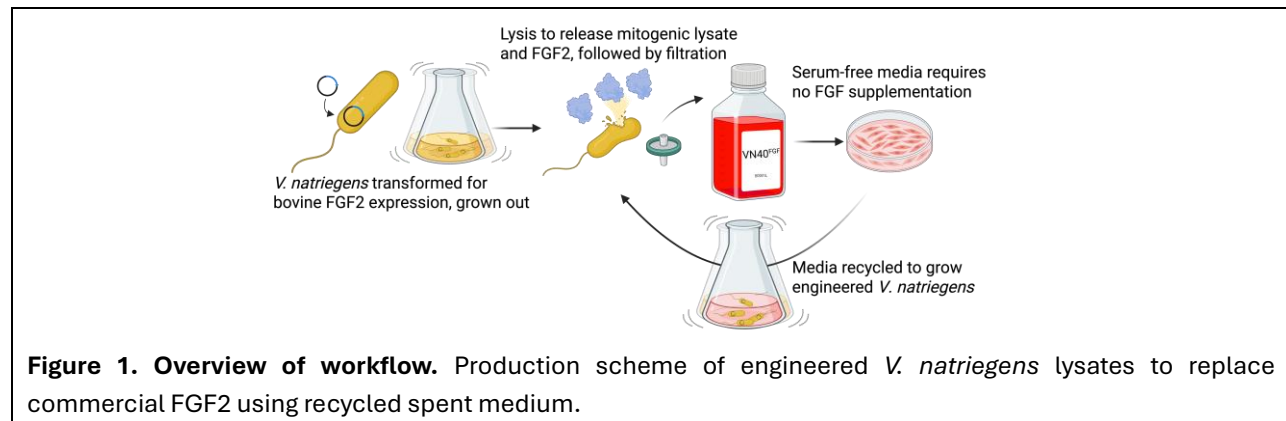


Figure 1. Overview of workflow. Production scheme of engineered *V. natriegens* lysates to replace commercial FGF2 using recycled spent medium.

V. natriegens NC7 can be engineered to express bovine FGF2.

We demonstrated the successful preparation of whole-cell lysate from *V. natriegens* NC7 strain engineered to express bovine FGF2 (bFGF2) using simple steps of growth, sonication, and filtration (**Figure 1**). Visualization of the total soluble protein content by SDS-PAGE showed the presence of a ~17 kDa band for the recombinant *V. natriegens*, suggesting successful expression of bFGF2 relative to the parental strain (**Figure 2A**). Whole-protein concentration of the lysate was ~3,800 $\mu\text{g}\cdot\text{mL}^{-1}$, with 66 $\mu\text{g}\cdot\text{mL}^{-1}$ (i.e., < 2 % total protein) being bFGF2 (**Figure 2B**). Thus, bFGF2 can be delivered to iBSCs at ~700 $\text{ng}\cdot\text{mL}^{-1}$ when including 40 $\mu\text{g}\cdot\text{mL}^{-1}$ *V. natriegens* lysate in media,¹⁶ which is significantly higher than the minimum 40 $\text{ng}\cdot\text{mL}^{-1}$ of hFGF2 required for robust growth in serum-free conditions.⁶

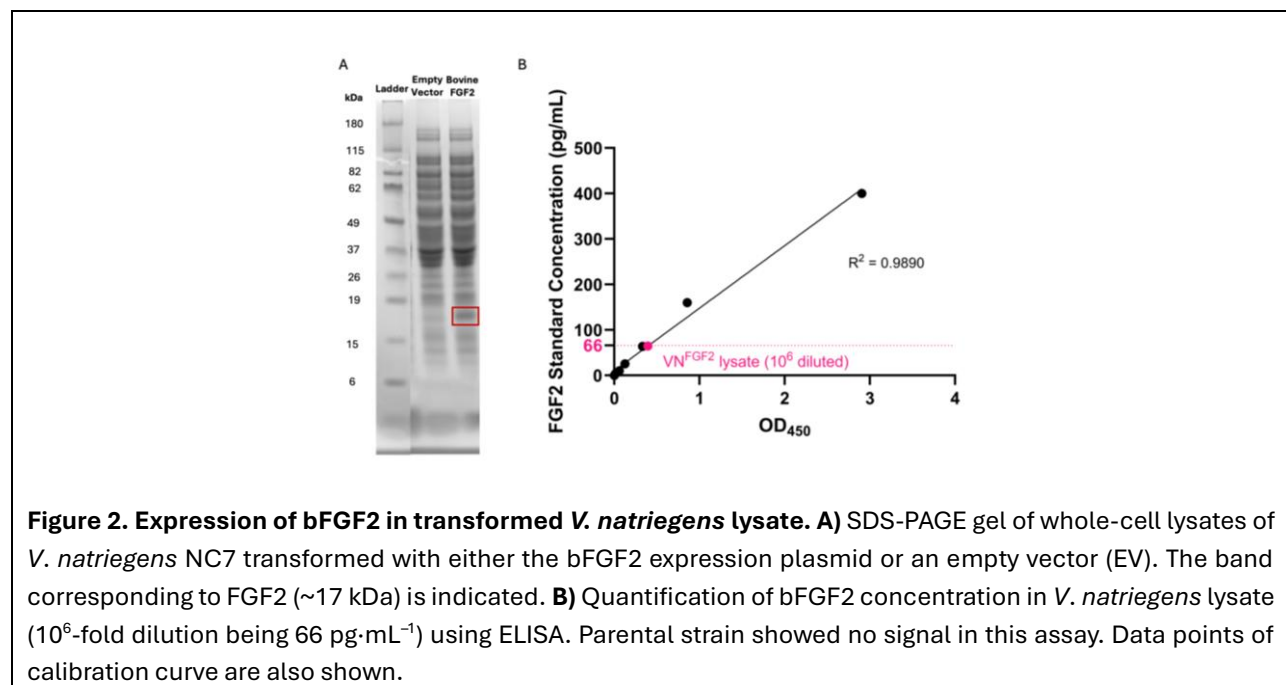
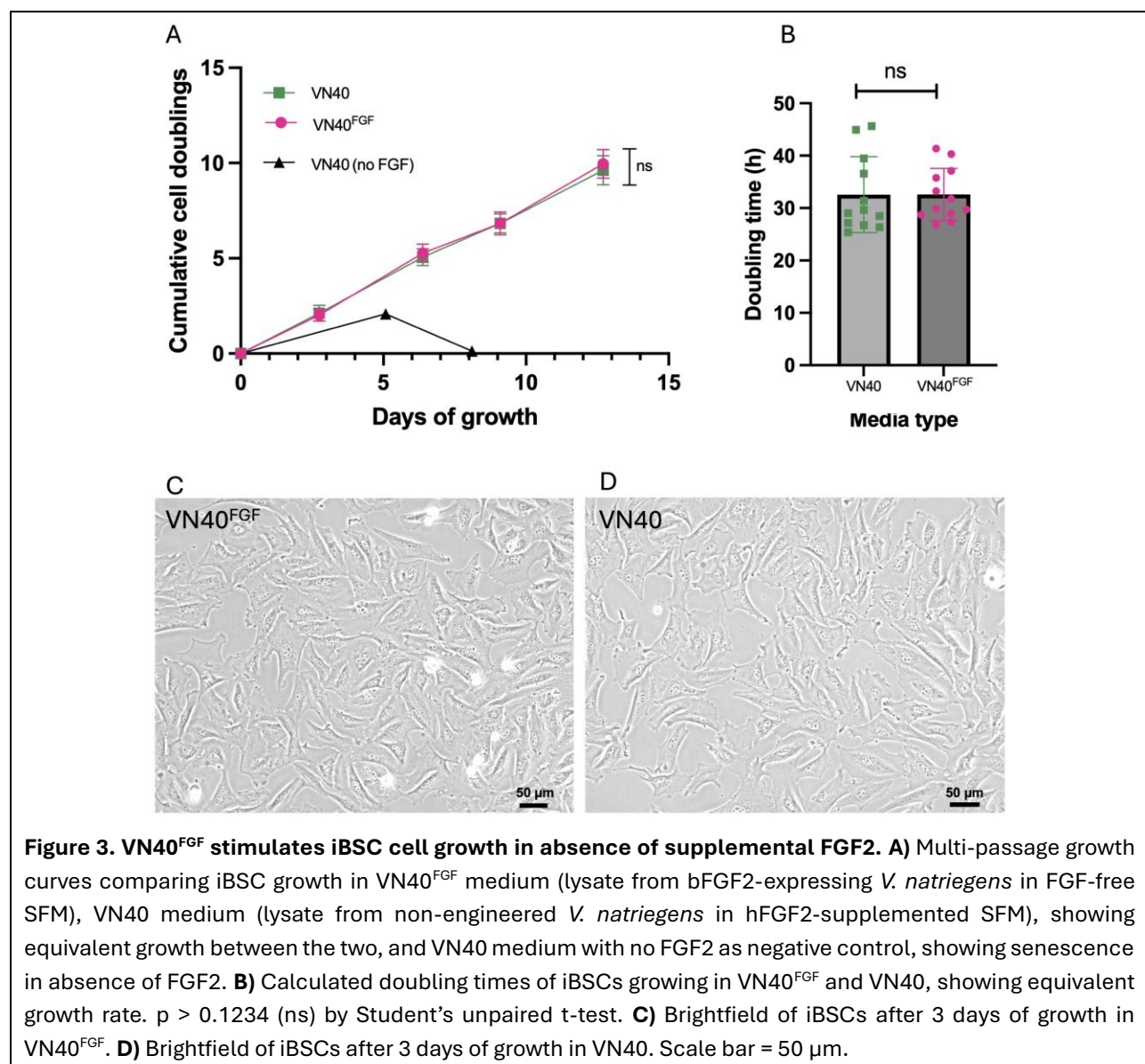


Figure 2. Expression of bFGF2 in transformed *V. natriegens* lysate. **A)** SDS-PAGE gel of whole-cell lysates of *V. natriegens* NC7 transformed with either the bFGF2 expression plasmid or an empty vector (EV). The band corresponding to FGF2 (~17 kDa) is indicated. **B)** Quantification of bFGF2 concentration in *V. natriegens* lysate (10⁶-fold dilution being 66 $\text{pg}\cdot\text{mL}^{-1}$) using ELISA. Parental strain showed no signal in this assay. Data points of calibration curve are also shown.

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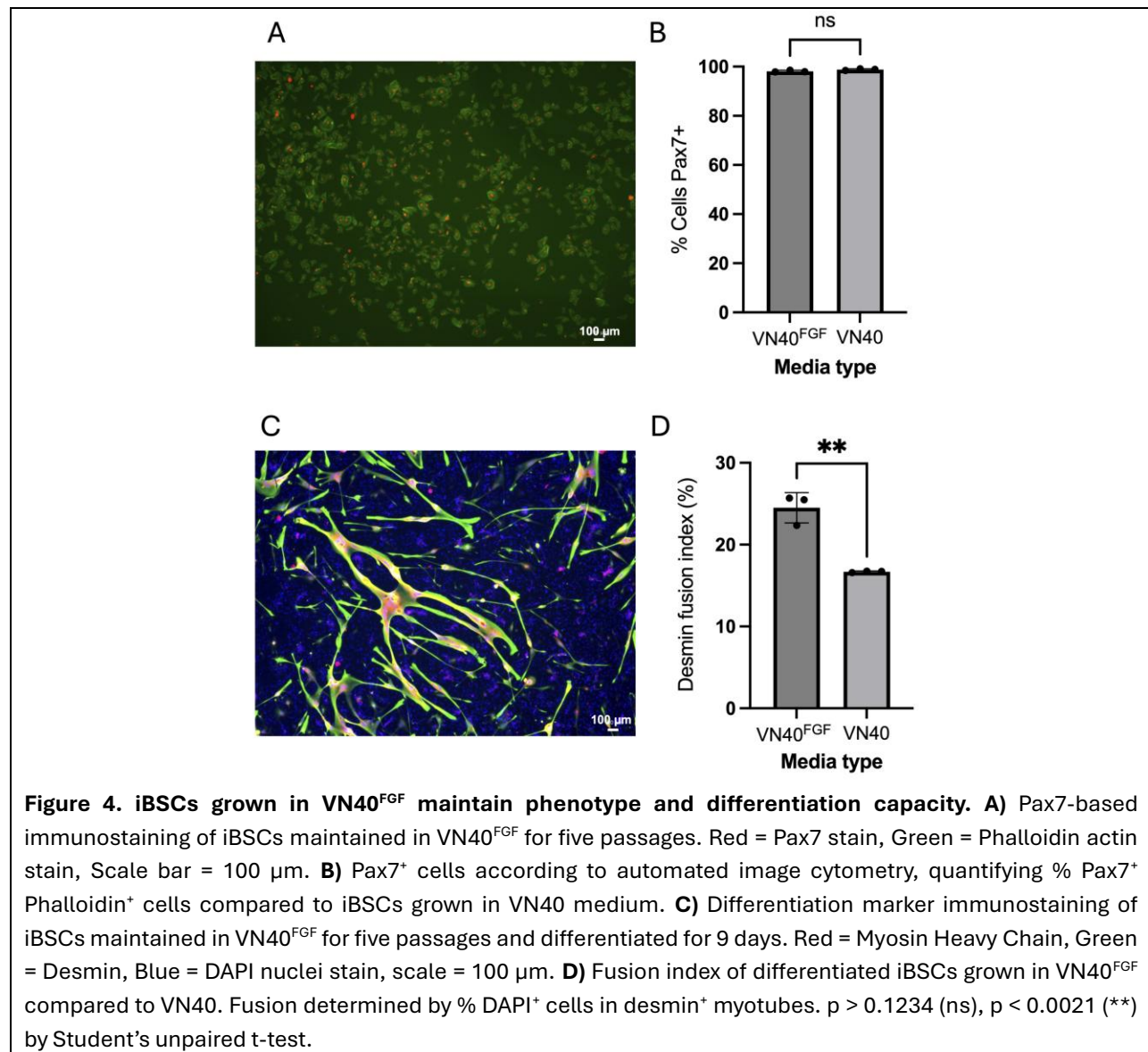
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VN40^{FGF} replaces FGF2 needs in VN40 for long-term iBSC proliferation.

We previously generated VN40 medium by adding lysate from wild-type *V. natriegens* at 40 μ g·mL⁻¹ protein and recombinant purified human FGF2 (hFGF2) at 40 ng·mL⁻¹ to B8 basal medium.¹⁶ To create VN40^{FGF}, we added lysate from bFGF2-expressing engineered *V. natriegens* at 40 μ g·mL⁻¹ protein to B8 medium. After adaptation to their respective media, we compared the growth kinetics of iBSCs in VN40^{FGF} and VN40 media. Cells grew at equivalent rates in both VN40^{FGF} and VN40, demonstrating the ability of expressed bFGF2 to replace supplemental hFGF2 in iBSC culture (**Figure 3A**). Doubling times were ~32.6 h in both media (**Figure 3B**), which was slower than those previously reported for serum-containing media (~22 h), consistent with prior studies with these cells.¹⁶ However, iBSCs

doubled more rapidly in VN40^{FGF} than in other FGF-replacement media, such as in recent attempts to engineer iBSCs to self-express bFGF2 (66 h).²⁶ We also confirmed cell morphology was comparable for cells grown in VN40^{FGF} (**Figure 3C**) and VN40 (**Figure 3D**). Thus the cells were not negatively impacted by bFGF2 expressed by *V. natriegens* or by the higher bFGF2 concentration delivered in VN40^{FGF}. These results demonstrate for the first time that recombinant growth factors expressed in the fast-growing *V. natriegens* can stimulate animal cell proliferation.



iBSCs grown in VN40^{FGF} maintain stemness and differentiation capacity.

Using Pax7 immunostaining, microscopy (**Figure 4A**), and quantitative image cytometry (**Figure 4B**) >98 % of the iBSCs maintained satellite cell phenotype over five passages in

VN40^{FGF}.²⁷ This was equivalent to cells passaged in standard VN40 medium, showing bFGF2 expressed from *V. natriegens* did not negatively impact stemness phenotype in the cells. After multiple passages in VN40^{FGF}, we were able to differentiate the iBSCs into mature multinucleated myotubes expressing Desmin and Myosin Heavy Chain proteins (**Figure 4C**), indicating early- and late-stage maturation, respectively.²⁸ These results demonstrate that cells were able to differentiate successfully despite the higher FGF2 concentration from extracts of VN40^{FGF}. In fact, cells grown in VN40^{FGF} differentiated with higher myotube formation than cells grown in VN40 (**Figure 4D**), perhaps due to higher concentrations of FGF2 impacting extracellular signal-regulated kinase (ERK) pathways and causing increased differentiation as shown previously.²⁹

***V. natriegens* lysate can be produced from spent cell culture media.**

To increase sustainably and circularity of the process, we grew engineered *V. natriegens* in spent VN40^{FGF} after iBSC growth. A potential obstacle in growing *V. natriegens* in spent cell culture medium is the presence of antibiotic-antimycotic (anti-anti), which is added to mammalian cell culture medium to prevent contamination. Treating the media with a 40 min autoclave cycle enabled *V. natriegens* growth, suggesting that autoclaving destroyed the antimicrobial components in the media. To investigate the limitations of DMEM/F12 media on *V. natriegens* growth, we measured *V. natriegens* growth over 20 h in LB, fresh VN40^{FGF}, and spent VN40^{FGF} (**Figure 5A**). All media conditions were supplemented with salt to mimic typical saline growth conditions of *V. natriegens*. VN40^{FGF} media, both spent and fresh, supported growth of *V. natriegens*, though rich media (LB + 2 % NaCl) yielded a greater maximum OD₆₀₀ (1.37 for LB vs. 0.5 for spent VN40^{FGF}).

To determine nutrient composition of spent VN40^{FGF} medium, we analyzed the metabolites,³⁰ and found increases in ammonium and lactate, a decrease in glutamine, and no change in glutamate, potassium and glucose (**Figure 5B**). After growing *V. natriegens* in spent media, we found almost complete consumption of glucose, glutamine, and lactate and significant production of ammonium. Thus, *V. natriegens* rapidly consumes the organic components of spent culture medium (likely glucose and glutamine first,³¹ followed by lactate).³² Spent VN40^{FGF} is therefore sufficiently nutrient-rich to support its use as recycled medium for *V. natriegens* growth, though growth could be improved through supplementing spent media with additional substrates. When growing *V. natriegens* in LB, we similarly observed ammonium accumulation (**Figure S2**), showing growth in both media are limited by the availability of carbon, not nitrogen.

The amount of bFGF2 present in *V. natriegens* lysate was quantified using ELISA. *V. natriegens* grown in spent medium yielded less bFGF2 than that grown in LB (212 vs. 700 ng·mL⁻¹ in VN40^{FGF}), which was consistent with their respective growth profile (**Figure 5C**).

However, both media conditions generated a large excess of bFGF2 compared to the amount of commercial FGF2 that is added to SFM ($40 \mu\text{g} \cdot \text{mL}^{-1}$). Using VN40^{FGF} generated from spent media resulted in iBSC growth similar to VN40^{FGF} generated from standard LB (**Figure 5D**), suggesting the activity of the lysate and recombinant growth factor is not negatively impacted by lower expression in the spent growth medium. We show that spent VN40^{FGF} medium pooled from three passages of iBSCs was sufficient to re-generate VN40^{FGF} for three additional passages. This shows an efficient media recycling process in which LB inputs can be minimized during manufacturing.

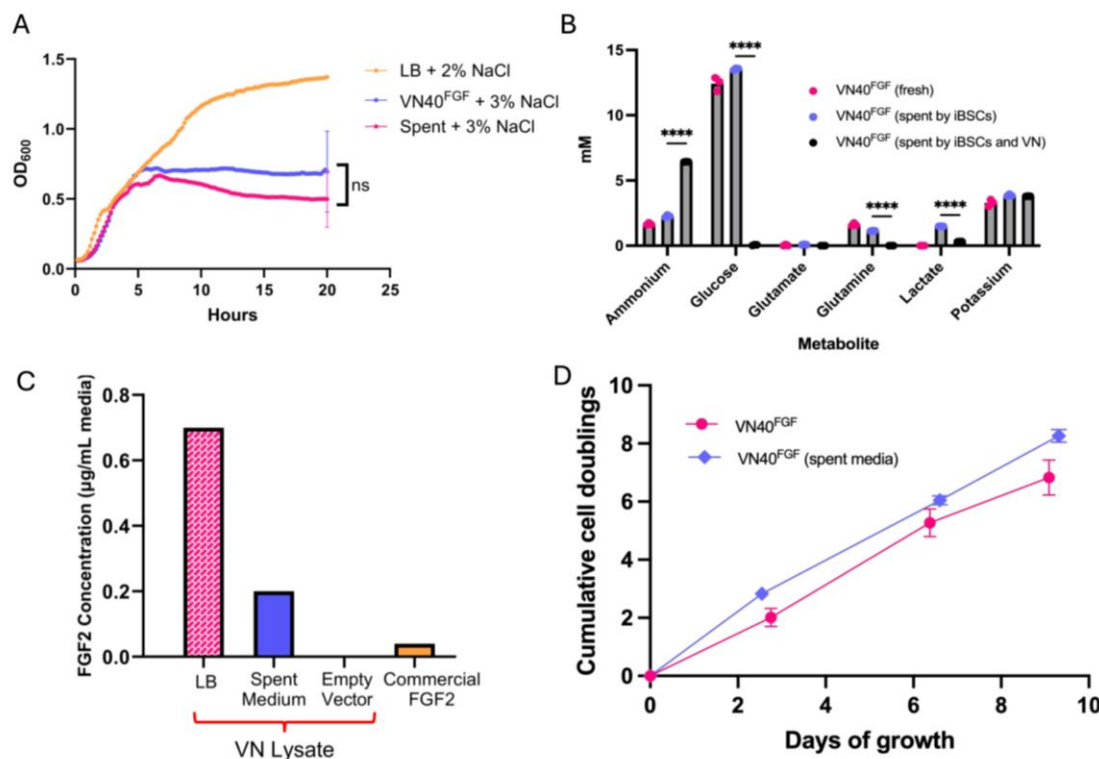
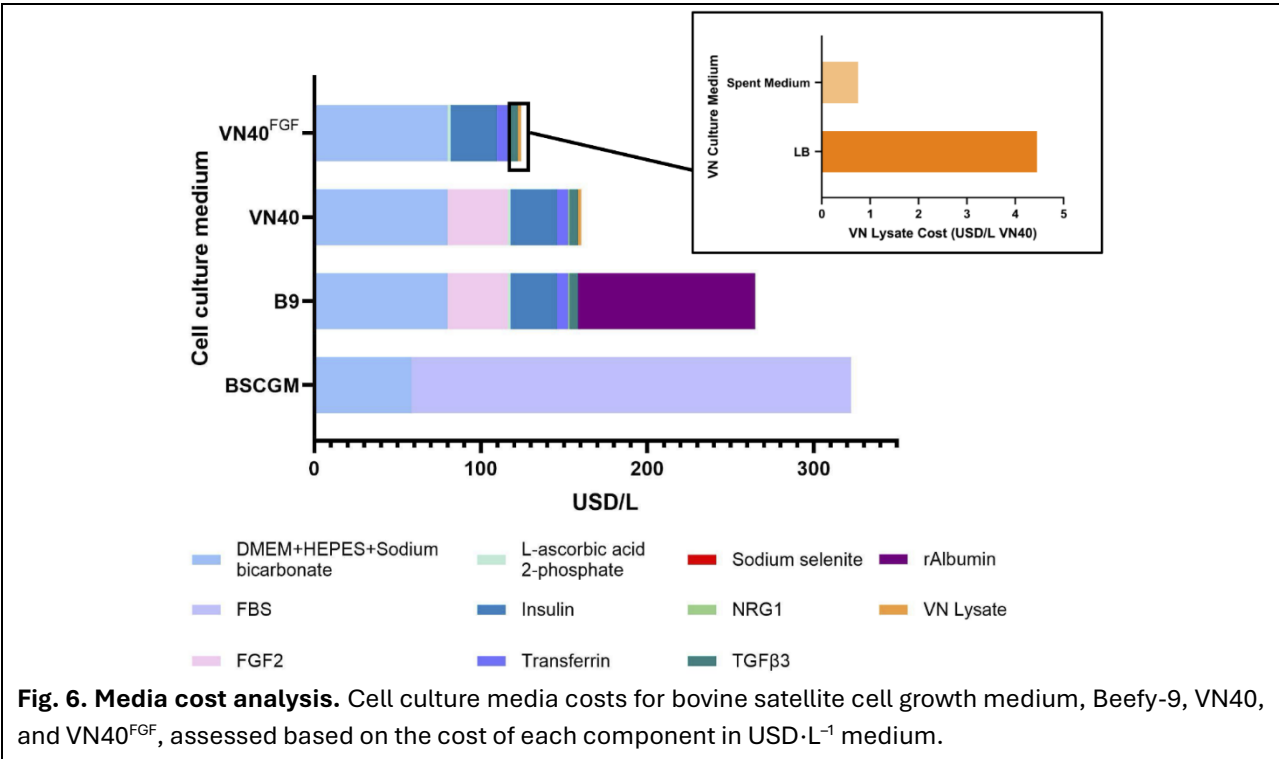


Figure 5. Spent medium recycling enables circular VN40^{FGF} lysate production. **A)** *V. natriegens* growth in salt-supplemented and autoclaved LB, fresh VN40^{FGF}, and spent VN40^{FGF}. One-way ANOVA with Dunnett's correction for multiple comparisons, $p < 0.05$ (ns). Data points shown are the mean of three replicates. **B)** Metabolite analysis of fresh VN40^{FGF}, spent VN40^{FGF} after growth of iBSCs (with media collection every 2–3 days), and spent medium after growth of *V. natriegens*. One-way ANOVA with Dunnett's correction for multiple comparisons, $p < 0.0001$ (****). **C)** Quantification of bFGF2 delivered in VN40^{FGF} by *V. natriegens* lysate grown in LB ($700 \text{ ng} \cdot \text{mL}^{-1}$) and spent medium ($212 \text{ ng} \cdot \text{mL}^{-1}$) compared to empty vector control (< detection limit) and concentration of commercial FGF2 added to SFM media. **D)** iBSC growth in VN40^{FGF} medium containing either lysate grown in LB or spent medium over 9 days ($n = 3$ replicates at each timepoint).

VN40^{FGF} produced using spent medium is cost-competitive in comparison to previous iBSC growth medium formulations.

To determine the cost-benefit of replacing commercial hFGF2 with our engineered lysate, we performed a analysis by tabulating the reagent costs of each media formulation iteration (**Figure 6**). The previous VN40 formulation had resulted in a significant cost reduction with respect to serum-containing bovine satellite cell growth medium (BSCGM), reducing media cost from \$323 L⁻¹ to \$158 L⁻¹, a 51 % reduction (**Table S1–S3**).¹⁶ By replacing hFGF2 in the media with whole-cell lysate, we further lower the media cost to \$124 L⁻¹, achieving a further 22 % cost reduction with respect to VN40 and a 62 % reduction with respect to BSCGM (serum-containing media) (**Table S3–S4**). Next, we determined the cost benefit of growing *V. natriegens* in recycled spent medium rather than LB. The reagent costs to produce *V. natriegens* lysate using LB equate to \$4.50 L⁻¹ of VN40, but by growing the bacterium in spent medium, we reduce this cost by 72 % to \$1.25 L⁻¹ (**Table S5–S6**). Importantly, these costs represent reagents purchased for lab-scale production. Media costs are expected to fall several orders of magnitude with large-scale amino acid production, shifts away from pharmaceutical-grade materials, and further decreases to growth factor costs.³³



Discussion

Cultivated meat offers a potential solution to mitigate the environmental impact of livestock generated meat for consumption. However, this process must be cost-effective and sustainable to achieve widespread accessibility and appeal. The resource burden of cell culture media is a significant roadblock to this goal.^{3,34} The high cost and environmental impact of media for cultivated meat is largely attributed to the use of fetal bovine serum (FBS) and intensive growth factor purification processes.^{4,8} Simplifying growth factor preparation can lead to significant reduction in resource needs for media production. Recently, we developed VN40 as an SFM using nutrient-rich whole-cell lysates from the rapidly dividing *V. natriegens* bacterium.¹⁶ This medium was advantageous, as it did not involve lengthy extraction or purification processes,^{9,17,19,21,35-36} instead requiring only a simple process of mechanical lysis and filtration to produce serum-replacing components. Still, VN40 contains purified, recombinant FGF2, which contributes significant cost. Further, the media used to grow *V. natriegens* may contribute to increased costs and environmental impact at scale.³⁷⁻³⁸ Our work shows that by engineering *V. natriegens* to express bFGF2, we can generate whole-cell lysates to replace both serum and hFGF2 in cultivated meat production without resource-intensive expensive purification processes. To further reduce the costs of *V. natriegens* culture, we also showed that these lysates can be produced circularly using spent media from mammalian cell culture.

V. natriegens has been used primarily for production of therapeutic proteins,²³ but here we also demonstrate its production of a recombinant growth factor. Growth factors like FGF2 and insulin-like growth factor 1 (IGF-1) used for cultivated meat production are typically produced in recombinant *E. coli*.¹⁰ *V. natriegens* possesses several potential advantages over *E. coli* for growth factor production – such as improved protein folding,³⁹ reduced endotoxin content,⁴⁰ and rapid growth rate.⁴¹ Combining this bacterium's efficient biomass production rate and recombinant expression abilities, we developed VN40^{FGF} from whole-cell lysates of bFGF2-expressing *V. natriegens*. With one liter of *V. natriegens* culture, we can produce 152 mg total lysate protein, enough for almost 4 L of cell culture media. While serum-free iBSC media, such as VN40, contains a controlled quantity of 40 ng·mL⁻¹ hFGF2, the VN40^{FGF} contained about VN40^{FGF} elicited iBSC growth comparable to VN40. This implies that bFGF2 from *V. natriegens* functions properly for binding and activation of bovine FGF receptors, and that iBSCs are tolerant of higher FGF2 concentrations than previously explored.⁹ Further studies could explore the dose-dependent potency of *V. natriegens* versus *E. coli*-expressed bFGF2. Further, total bFGF2 could be titrated to lower levels if needed via dilution with extracts from the untransformed *V. natriegens*. While we used a bovine FGF2, future studies can explore proteins sourced from other species as these could prove to be

more potent.⁹ The use of bovine FGF2 is advantageous over human FGF2 used in prior studies from a regulatory and consumer acceptance standpoint.^{6,16}

iBSCs demonstrated maintenance of satellite cell phenotype and differentiation capacity after growth in VN40^{FGF}, despite the overabundance of bFGF2 present in this medium. While FGF2 is primarily a cell cycle progressing factor, studies point to its role in stimulating muscle regeneration and differentiation,^{29,42} which could explain higher differentiation in VN40^{FGF}-fed cells. Evidently, residual bFGF2 remaining after the removal of proliferation medium did not cause issues for differentiation. This is cited as an issue in studies exploring engineering bovine cells for self-signaling bFGF2 production,²⁶ where its biosynthesis cannot be turned off, causing interference with differentiation. VN40^{FGF} also does not rely on engineered bovine cells, which could raise regulatory and consumer acceptance concerns. While we demonstrate that our lysate promoted iBSC growth and differentiation to an equal or greater extent as did commercial FGF2-containing media, we note that the stability and structure of our *V. natriegens*-generated bFGF2 has not been investigated. Under the cultivation conditions utilized here, it is sufficiently stable to have a positive biological impact.

We saw an additional opportunity to reduce resource burden of VN40^{FGF} production by cultivating *V. natriegens* on spent cell culture media rather than using fresh bacterial growth medium. This promises to significantly reduce overall cost and environmental impact of cultivated meat media.^{43,44} We fully repurposed our spent iBSC cell culture media for VN40^{FGF} growth, which was then used for equivalent cell growth. This demonstrates a fully circular process, where mammalian cell growth can subsist entirely on VN40^{FGF} produced from prior passages. While growth of *V. natriegens* on spent media was inferior to that in LB, sufficient lysate was produced for additional cycles of iBSC growth. *V. natriegens* growth in spent VN40^{FGF} did not differ with fresh VN40^{FGF}, though analysis of spent VN40^{FGF} media revealed an increase in ammonium and lactate, and a decrease in glutamine, compared to fresh VN40^{FGF}. These metabolite differences did not impact nutrient availability for *V. natriegens* to successfully grow, though higher densities of suspension mammalian cell culture may cause higher nutrient depletion. Prior spent media analyses suggest the rapid consumption of glutamine by mammalian cells, but a more delayed consumption of glucose,⁴⁵ potentially explaining the unchanged glucose content during frequent media changes. Nutrient availability for *V. natriegens* growth showed that the cell culture media could be further recycled through glucose and glutamine supplementation, which could also induce VN ammonium consumption.⁴⁶ Our results showed *V. natriegens* may be used for lactate removal to regenerate media for mammalian cell culture, which would recycle media and produce lysate simultaneously.

This work results in 22 % cost savings compared to VN40, and 62 % cost-savings compared to serum-containing media. Growth factors represent a significant cost in cultivated meat, and could comprise a greater share of cost at scale compared to other media ingredients.⁴⁷ Basal media represents the next most expensive media ingredient, though these costs are expected to drop dramatically upon scale.⁴⁸ Expression of other growth factors, such as TGFβ3 and transferrin, in engineered lysates could further decrease costs of media especially at scale. Processing improvements for lysate production, such as high-pressure homogenization instead of sonication, would further decrease costs at scale.³⁷ This work also results in significant decreases in resource use for cultivated meat. Growth factor purification involves resource-intensive column separation and dialysis, the need for which this work eliminates.⁸ Further resource demands and environmental externalities are caused by the fermentation step of growth factor production.⁸ Here, we reduce these burdens by demonstrating growth of engineered *V. natriegens* in spent media. Resource demand can be further reduced by taking advantage of ambient temperature fermentation.²⁴ Overall, VN40^{FGF} shows a low-cost and low-resource usage approach to creating SFM for cultivated meat that does not require supplemental growth factor addition. This work brings cultivated meat closer to reaching affordability and sustainable production.

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Conflict of Interest

Tufts University has applied for a patent on this work.

Author Contributions

James Dolgin: Methodology, Formal analysis, Investigation, Writing - Original Draft, Review & Editing, Visualization, Avani Vaid: Methodology, Formal analysis, Investigation, Writing - Original Draft, Review & Editing, Visualization, Dylan Hendrixon: Investigation, Yiming Cai: Investigation, Lennie K. Y. Cheung: Investigation, Licheng Xu: Investigation, David L. Kaplan: Writing - Review & Editing, Supervision, Funding acquisition, Nikhil U. Nair:

Conceptualization, Methodology, Writing - Review & Editing, Supervision, Funding acquisition.

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